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Research Article
doi.org/10.1002/ejoc.202401234

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Ni-Catalyzed Synthesis of Multisubstituted Allenes via Reductive Cross-Coupling of Propargylic Acetates with Chlorohydrosilanes

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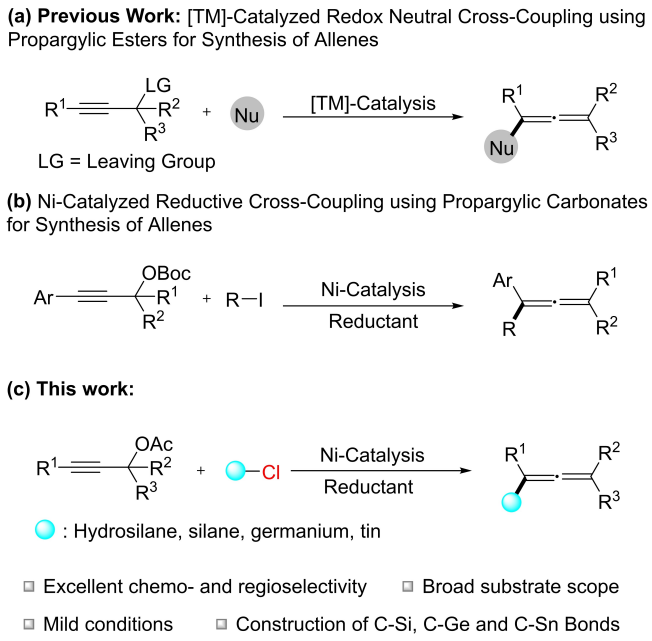
This paper introduces an innovative strategy for the synthesis of multisubstituted allenes via nickel-catalyzed reductive cross-coupling reactions. The approach utilizes propargyl acetates and chlorosilanes to produce a variety of silyl-substituted allenes. The reaction conditions are notably mild, and the process is characterized by high chemo- and regioselectivity, as well as an impressive substrate scope. Furthermore, the method can be readily extended to the use of chlorogermane and chlorostannane, facilitating the synthesis of germanium or stannium-substituted allenes. This research not only offers a valuable strategy for the preparation of multisubstituted allenes but also significantly broadens the synthetic repertoire for the construction of highly functionalized molecules.

Introduction

Allenes are pivotal molecular structures that are integral to a myriad of natural products and bioactive pharmaceuticals.[1] Their versatile reactivity positions allenes as indispensable components in the synthesis of natural products and pharmaceutical compounds.[2] As a result, allenes are unmatched in the realm of organic synthesis, making their synthesis a topic of considerable interest among chemists. The broad applicability of allenes is due to their unique chemical structure, which is crucial in organic synthesis, natural product research, pharmaceutical development, and the creation of functional materials.[3] Currently, various strategies for the synthesis of allenes have been established,[4] including the transition metal-catalyzed nucleophilic substitution reactions of propargyl compounds containing leaving groups (Scheme 1a).[5] These reactions can utilize a range of reactive groups such as acetates, methanesulfonates, carbonates, and halogens, as leaving groups. Grignard reagents, organoborons, terminal alkynes, hydrosilanes, and other nucleophilic reagents have been demonstrated to be effective coupling partners.[6] The recent progress in nickel-catalyzed reductive cross-coupling process has introduced a novel approach to the synthesis of allenes. Wang's group has successfully synthesized multisubstituted allenes using propargylic carbonates and alkyl iodides in a nickel-catalyzed reductive cross-coupling process (Scheme 1b).[7]

Organosilanes, with their extensive applications in pharmaceuticals, pesticides, and functional materials, have sparked significant research interest.[8] Traditional methods for constructing C(sp^2)–Si bonds typically involve cross-coupling reactions with organometallic reagents or hydrosilanes and halides.[9] Shu's group recently achieved successful nickel-catalyzed cross-electrophilic coupling of aryl or alkenyl halides with chlorosilanes.[10] Subsequently, other research groups also developed several examples of nickel-catalyzed reductive cross-coupling of chlorosilanes to form C–Si bonds.[11]

A novel nickel-catalyzed reductive cross-coupling reaction has been successfully developed for the synthesis of multisubstituted silylallenes (Scheme 1c). This reaction entails the interaction of chlorohydrosilanes with propargylic acetates. Moreover, the versatile reaction conditions can be adapted for the synthesis of alkylgermane-substituted allenes and alkyltin-



Scheme 1. Transition metal-catalyzed cross-coupling using propargyl compounds for synthesis of allenes.

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Supporting information for this article is available on the WWW under https://doi.org/10.1002/ejoc.202401234

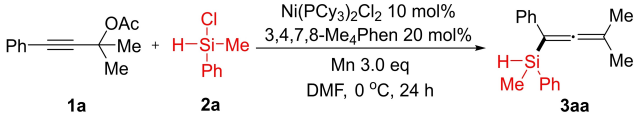
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substituted allenes. Recently, Li's group reported instances of nickel-catalyzed reductive coupling for the synthesis of silylenes.[12] Additionally, there have been reports on the synthesis of silylenes through electroreductive cross-coupling.[9f],[9i] However, existing examples have focused solely on radicals, whereas our reaction opens up new possibilities by achieving reductive cross-coupling with chlorohydrosilanes, allowing for a wider range of substrate derivatisation in later stages.

Results and Discussion

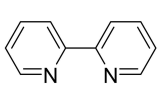
Initial investigations into the synthesis of allenes from propargyl acetate **1a** and chloromethylphenylsilane **2a** via nickel-catalyzed reductive cross-coupling have been conducted, with the outcomes detailed in Table 1. Through a systematic optimization process, it was determined that employing $Ni(PCy_3)_2Cl_2$ as the catalyst, 3,4,7,8-Me_4Phen as the ligand, and Mn as the reducing agent, the desired silyl-substituted allene **3aa** could be synthesized with an 89% yield after a 24-hour reaction at 0 °C in DMF (entry 1). In contrast, the commonly utilized catalysts $Ni(COD)_2$ and $NiCl_2$ yielded only 53% and 38% of product **3aa**, respectively (entries 2–3). It is hypothesized that PCy_3 in $Ni(PCy_3)_2Cl_2$ may also function as a ligand within the catalytic cycle. However, the introduction of bpy as a ligand resulted in a significant decrease in reaction yield (entry 4). The influence of alternative reducing agents was also examined, with zinc powder leading to a diminished yield (entry 5). Furthermore, the solvent's impact on the reaction was scrutinized, with THF showing a marked reduction in yield (entry 6). Reducing the amount of chloromethylphenylsilane **2a** to equimolar proportions resulted in a 53% yield decrease (entry 7). Additionally, abbreviating the reaction time to 15 hours was detrimental, as the starting material was not completely consumed, leading to a lower yield (entry 8). Moreover, conducting the reaction at room temperature significantly impaired the yield of product **3aa** (entry 9). It is crucial to emphasize that both the Ni catalyst and the reducing agent Mn are indispensable for the reaction; without their presence, the product **3aa** cannot be obtained (entry 10).

Table 1. Optimization of the reaction conditions.[a]

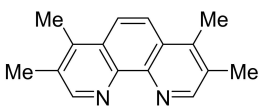


1a **2a** **3aa**

Entry	Variations from standard conditions	Yield of **3aa**(%)[b]
1	none	89 (82)[c]
2	$Ni(COD)_2$ instead of $Ni(PCy_3)_2Cl_2$	53
3	$NiCl_2$ instead of $Ni(PCy_3)_2Cl_2$	38
4	bpy instead of 3,4,7,8-Me_4Phen	7
5	Zn instead of Mn	68
6	THF instead of DMF	41
7	Add 0.40 mmol of **2a**	53
8	15 h instead of 24 h	75
9	At room temperature	63
10	Without Ni or Mn	0



bpy



3,4,7,8-Me_4Phen

[a] Reaction conditions: **1a** (0.20 mmol), **2a** (0.60 mmol), DMF (1.5 mL), under N2. [b] Yield determined by gas chromatography analysis using dodecane as internal standard. [c] Isolated yields.

Having optimized the reaction conditions for the reductive coupling reaction to synthesize silyl-substituted allenes, we proceeded to explore the versatility of this method using a variety of propargyl

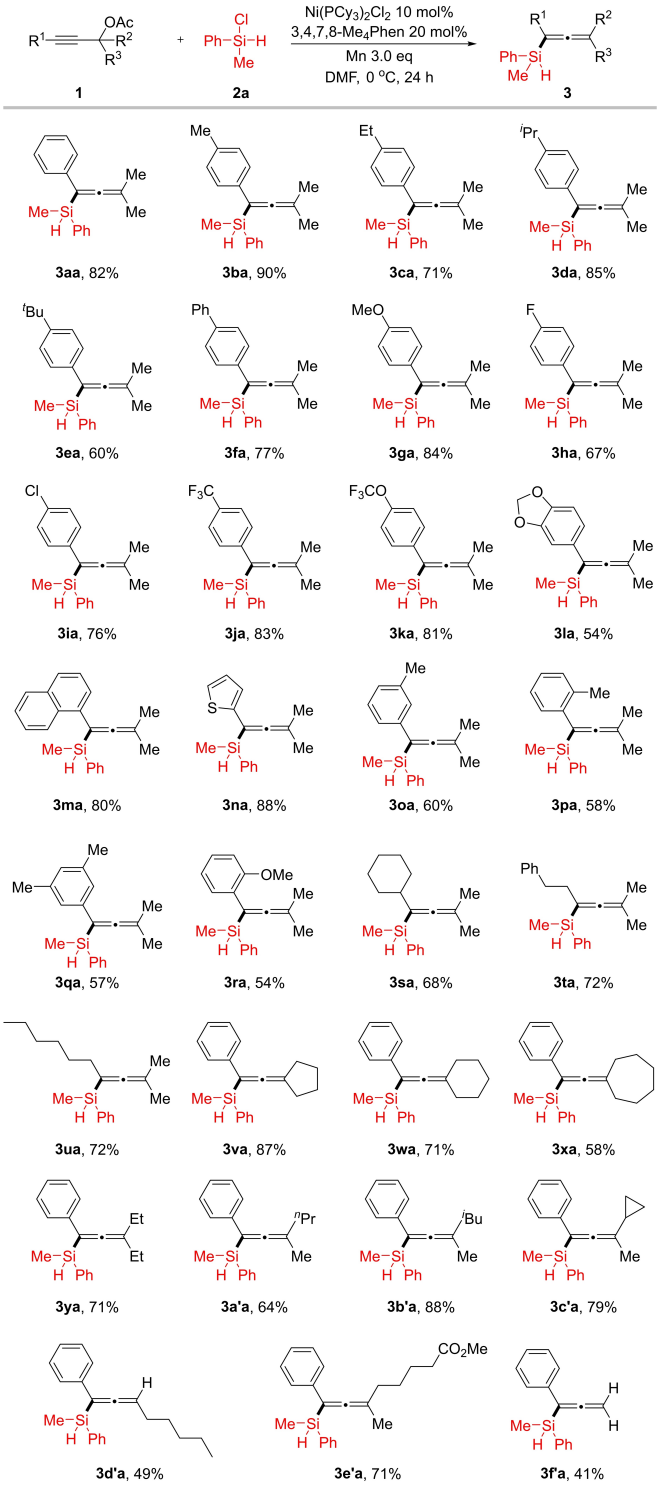
acetates with different structural and functional groups (Scheme 2). Reassuringly, > arylpropargyl acetates with para-alkyl substituents were found to be highly amenable to the reaction, yielding the corresponding allene products (**3ba-3ea**) with 60–90% yields. The reaction was also efficient with arylpropargyl acetates bearing phenyl substituents, such as **1f**, which produced the product **3fa** in good yields. Furthermore, arylpropargyl acetates containing electron-donating methoxyl groups were compatible with the C–Si reductive cross-coupling, affording the target molecule **3ga** in an 84% yield. Substrates with electron-withdrawing groups like F, Cl, CF3, and OCF3 also performed well, yielding products **3ha-3ka** in the range of 67%-83% yields. Furthermore, the reaction demonstrated broad compatibility with a range of substrates, including those featuring 1,3-benzodioxole, 1-naphthalene, thiophene, and *m*-Me groups. These diverse substrates were successfully transformed into their respective allene products (**3la–3oa**) with yields ranging from 54% to 88%. Notably, substrates **1q** and **1r**, featuring 3,5-dimethyl and o-methoxy substitutions, respectively, successfully yielded their corresponding products. It is significant to highlight that this reaction is not limited to aryl-substituted substrates, as alkyl-substituted substrates, including cycloalkanes and chain alkanes (**1s-u**), were also found to be compatible. These alkyl-substituted substrates enabled the efficient synthesis of the corresponding allenes with high yields. This finding was further supported by the successful

compatibility of arylpropargyl acetate substrates with a cycloalkane skeleton. We tested cyclopentane, cyclohexane, and cycloheptane derivatives, all of which smoothly underwent the reaction to yield the desired products **3va–3xa**. Additionally, when the substrate was switched to two ethyl-substituted phenylpropargyl acetates, the product **3ya** was obtained in a 71% yield. Lastly, we investigated unsymmetrical substrates (**1a′–f′**) and achieved favorable results under the established reaction conditions.

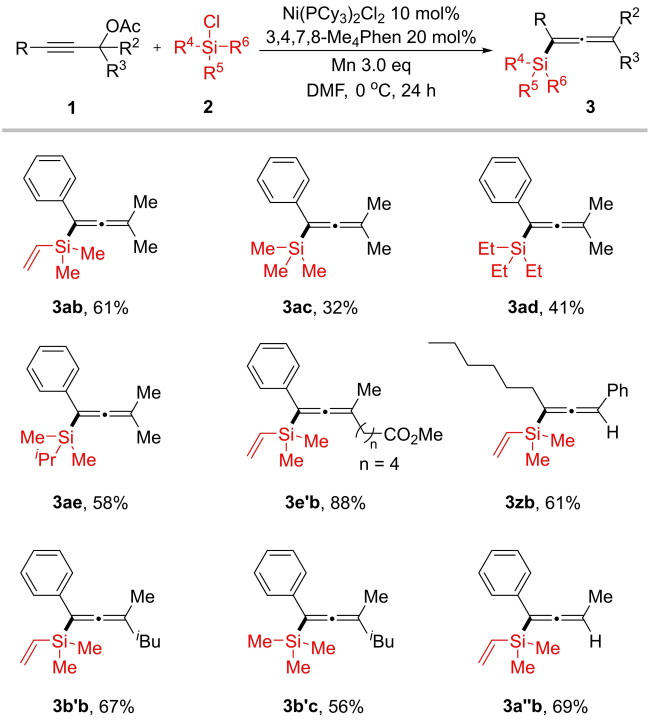
Following an in-depth investigation into the versatility of propargyl acetate substrates in nickel-catalyzed reductive cross-coupling reactions, we redirected our focus to assess the suitability of a variety of chlorosilanes as potential coupling partners in this reaction (Scheme 3). Our studies revealed that not only hydrochlorosilanes but also other chlorosilanes are amenable to this reaction. For example, vinyl dimethyl chlor-

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Scheme 2. Substrate Scope.[a]
[a] Reaction conditions: **1** (0.20 mmol), **2a** (0.60 mmol), $Ni(PCy_3)_2Cl_2$ (10 mol %), 3,4,7,8-Me_4Phen (20 mol %), Mn (3.0 equiv), DMF (1.5 mL), 24 h, under N_2.



Scheme 3. Scope of chlorosilanes.[a]
[a] Reaction conditions: **1** (0.20 mmol), **2** (0.60 mmol), $Ni(PCy_3)_2Cl_2$ (10 mol %), 3,4,7,8-Me_4Phen (20 mol %), Mn (3.0 equiv), DMF (1.5 mL), 24 h, under N_2.

osilane **2b** was successfully reacted to yield allene products. Additionally, trimethylchlorosilane, a staple reagent in organic synthesis, proved to be compatible with the reaction, leading to the synthesis of the trimethylsilyl-substituted allene **3ac**. Triethylchlorosilane also yielded similar results, and we further explored the reaction's tolerance to dimethylisopropylchlorosilane, which afforded the target molecule **3ae** in moderate yields. We then extended our investigation to the compatibility of vinyl dimethyl chlorosilane with a range of propargyl acetates. Moreover, the reaction's compatibility with unsymmetrical substrates was also scrutinized. It was observed that phenylpropargyl acetate substrates, which contain a chain methyl ester, are highly compatible with the reaction, yielding the desired product **3e'b**. Similarly, propargyl acetate substrates with long alkyl chain substitutions also produced the product **3zb** in good yields. For instance, the reaction of phenylpropargyl acetates with both methyl and isobutyl substitutions with vinyl dimethylchlorosilane and trimethylchlorosilane was successful, resulting in the formation of silyl-substituted allenes (**3b'b–3b'c**). It is noteworthy that phenylpropargyl acetate substrates with a single methyl substitution also performed well under the reaction conditions described, yielding satisfactory amounts of the desired product **3a''b**. This demonstrates the reaction's tolerance towards these unsymmetrical substrates, thus opening up new avenues for future developments in enantioselective synthesis of silyl-substituted allenes.

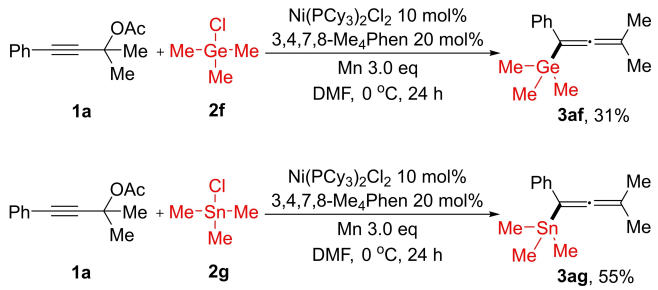
After delving into the nickel-catalyzed reductive coupling of propargyl acetates with chlorosilanes for the synthesis of allenes, we then set our sights on exploring the potential for coupling reactions with other elements from Group 14 of the periodic table (Scheme 4). Our ambition was to synthesize germanium-substituted allenes, and we achieved this by reductively coupling trimethylgermanium chloride with propargyl acetate under standard conditions. The resulting product,

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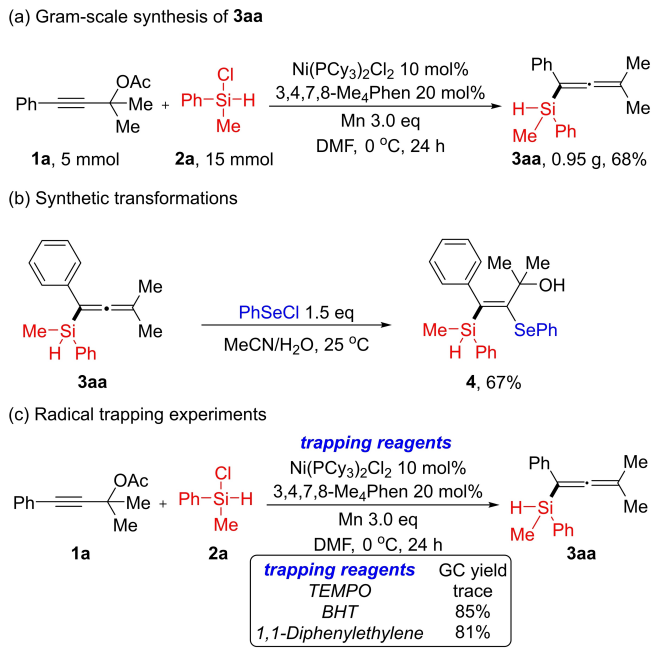
Scheme 4. Scope of chlorotrimethylgermane and trimethyltin chloride.[a]
[a] All reactions performed with **1a** (0.2 mmol) under standard reaction conditions.

3af, offers a direct route to the synthesis of germanium-substituted allenes, albeit with a relatively modest yield. We also contemplated whether the reductive coupling using trimethyltin chloride could be accomplished under the current reaction conditions. Fortunately, the alkyltin-substituted allene compound **3ag** was successfully synthesized within our reaction system, overcoming the synthetic challenges associated with conventional methods. The efficient and effective synthesis of multisubstituted allenes was thus realized through the successful synthesis of alkylgermanium- and alkyltin-substituted allenes.

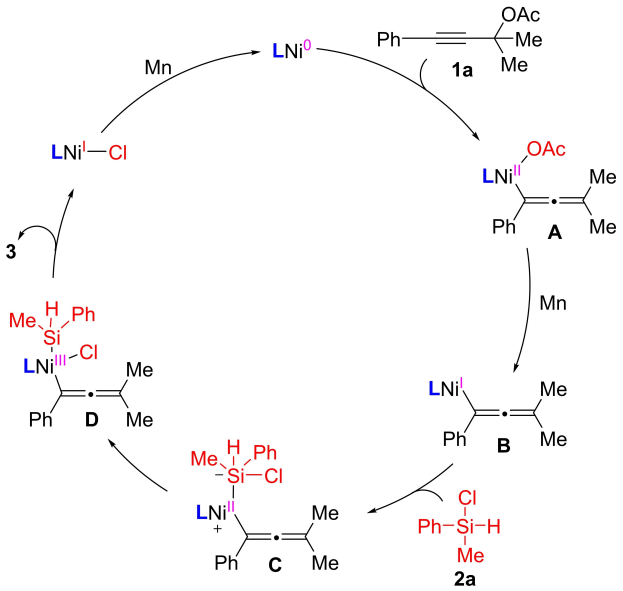
To verify the efficiency of the reaction, we also carried out a gram-scale synthesis, which gave 0.95 g of the allene product 3aa in 68% yield under standard reaction conditions (Scheme 5a). Following this, we initiated derivatisation studies on the silyl-substituted allene **3aa**, as depicted in Scheme 5b. The synthesis of tetrasubstituted vinyl product **4** was successfully achieved by treating the tetrasubstituted allene **3aa** with phenylselenium chloride. The reaction conditions were simple and accessible, facilitating the formation of compounds with four distinct substituents, including selenium and silyl groups. In an effort to deepen our understanding of the reaction mechanism, we conducted several experiments. To determine if the reaction proceeded via a radical pathway, we introduced radical inhibitors, such as (2,2,6,6-tetramethylpiperidin-1-yl)oxyradical (TEMPO), butylhydroxytoluene (BHT), and 1,1-diphenylethene into the reaction system, as illustrated in Scheme 5c. The results indicated that the presence of TEMPO significantly inhibited the reaction, while BHT and 1,1-diphenylethylene had no such inhibitory effect. The incorporation of TEMPO led to a substantial accumulation of propargyl acetate feedstock, which can be attributed to TEMPO's oxidizing properties that impede the initiation of the reaction.

Notably, the absence of captured radical intermediates suggests that the reaction does not follow a radical mechanism, further elucidating the non-radical nature of the reaction pathway.

The proposed catalytic cycle for the reductive cross-coupling silylation reaction, which facilitates the construction of multisubstituted allenes, is delineated in Scheme 6. This mechanism is corroborated by experimental data and aligns with findings from the existing literature. [13] The catalytic cycle commences under reducing conditions with the generation of a Ni(0)L species. The reaction is initiated by the oxidative addition of propargyl acetate **1a**, resulting in the formation of the allene Ni(II)L complex **A**. Following this, the introduction of the reducing agent Mn expedites the reduction of complex **A**, leading to the allene Ni(I)L intermediate **B**. The reaction sequence continues with a second oxidative addition involving chlorosilane **2a** and intermediate **B**, culminating in the formation of the silyl-substituted allene **3aa**. This product can be readily liberated by reduction from the ensuing allene Ni(III)L complex **D**. Concurrently, the Ni(I)L–Cl species is released. Ultimately, the Ni(0)L species is regenerated through a



Scheme 5. Gram-scale, synthetic transformations and mechanism study.



Scheme 6. Possible reaction mechanism.

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manganese-mediated reduction of Ni(I)L–Cl, thereby perpetuating the catalytic cycle.

Keywords: Ni-catalyzed · Reductive cross-coupling · Multisubstituted allenes · Propargylic acetates · Chlorohydrosilanes.

Conclusions

In conclusion, this study presents an innovative approach for the synthesis of silyl-substituted allenes through nickel-catalyzed reductive cross-coupling reactions between propargylic acetates and chlorohydrosilanes. This reaction system demonstrates versatility by accommodating a variety of chlorohydrosilanes and chlorosilanes, while showing exceptional tolerance towards different functional groups. The simplicity and mild conditions of this method make it a convenient pathway to access a diverse range of silyl-substituted allene compounds. Moreover, the broad applicability of this reaction strategy extends to chlorogermananes and chlorostannanes, thereby broadening the scope of possible reactions. Currently, our research efforts are directed towards exploring the enantioselective synthesis potential of this reaction, which could further enhance its synthetic utility and applicability in the field of organic chemistry.

Experimental Section

Typical Procedure for the Synthesis of Multisubstituted Allenes Compounds (3)

Procedure: The procedure was conducted in an nitrogen-filled glove box. To a reaction tube equipped with a magnetic stir bar was charged with $Ni(PCy_3)_2Cl_2$ (13.8 mg, 0.02 mmol), 3,4,7,8-Me_4Phen (9.4 mg, 0.04 mmol), Mn (33.0 mg, 0.6 mmol), propargyl ester **1** (0.2 mmol), Chlorosilanes **2** (0.6 mmol), and DMF (1.5 mL). The reaction tube was sealed and removed from the glove box. The reaction mixture was stirred at 0 °C for 24 h. The reaction was quenched with water (2.0 mL) and extracted with ethyl acetate (3 ×15.0 mL). The combined organic layers was washed with water (2 ×15.0 mL), brine (15.0 mL), dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The residue was purified by flash chromatography on silica gel (PE/EA, from 200:1 to 50:1) to afford the desired product **3**.

Acknowledgements

We thank the Natural Science Foundation of Anhui Education Department (2024AH051689) for financial support.

Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

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Manuscript received: November 5, 2024
Revised manuscript received: December 6, 2024
Accepted manuscript online: December 11, 2024
Version of record online: December 20, 2024

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